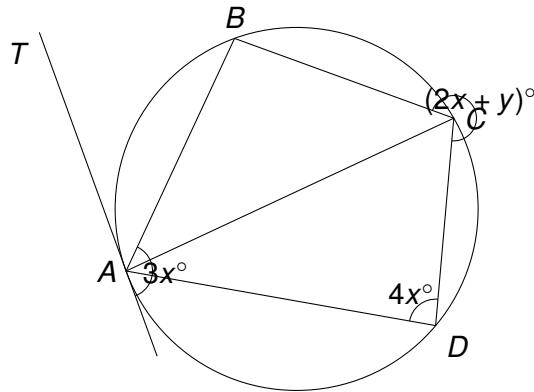


1 A , B , C and D are points on a circle with centre O . TA is a tangent to the circle at A .

$ABCD$ is a cyclic quadrilateral. Angle $BAT = 3x^\circ$, angle $BCD = (2x + y)^\circ$, and angle $ADC = 4x^\circ$.



Show that $y = x$.

You must show each stage of your working, naming any circle theorems you use.

(4)

(Total for Question 1 is 4 marks)
